Monotic Scheduling.

```
# include <stdio.h>

Struct task {
  int id, cpu, period;
};

Void monotic_scheduling(struct task t[], int n, int max_time){
  for (int i=0; i<n; i++){
     printf(" Task %d:\n", i+1);
     printf(" Id: ");
     Scanf("%d", &t[i].id);
     printf (" cpu time:");
     scanf("%d", &t[i].cpu);
     printf (" Period: ");
     scanf("%d", &t[i].period);
  }

  for (int i=0; i<n; i++){
     for (int j=0; j<n-i-1; j++){
        if (t[j].period > t[j+1].period){
           struct task temp = t[j];
           t[j] = t[j+1];
           t[j+1] = temp;
        }
     }
  }
  int remaining_time [n];
  int next_arrival[n];
  for (int i=0; i<n; i++){
     remaining_time [i] =0;
     next_arrival [i] =0;
  }
  printf ("\nGantt chart:\n");
  for (int time=0; time < max_time; time++){
     for(int i=0; i<n; i++){
        if (time == next_arrival [i]){
           remaining_time[i] = t[i].cpu;
           next_arrival[i] += t[i].period;
  }}
```

```
int current_task = -1;
for (int i=0; i<n; i++){
    if(remaining_time[i]>0){
        current_task = i;
        break;
    }
}
if (current_task != -1){
    ~~printf("[%.2d - %d] T%d \n", time, time~~
    printf("T%d", t[current_task].id);
    remaining_time[current_task]--;
}else{
    printf("Idle");
}
}
printf("\n");
}
int main(){
    int n, max_time;
    printf("Enter number of tasks:");
    scanf("%d", &n);
    Struct task t[n];

    for (int i=0; i<n; i++){
        printf("\nTask %d :\n", i+1);
        printf("Enter ID:");
        scanf("%d", &t[i].id);
        printf("Enter CPU time:");
        scanf("%d", &t[i].cpu);
        printf("Enter period:");
        scanf("%d", &t[i].period);
    }
    printf("\n Enter max time:");
    scanf("%d", &max_time);

    monotonic_scheduling (t, n, max_time);

    return 0;
}
```

~~EAT~~

O/P:

Enter number of tasks: 3

Enter simulation time: 20

task 1:

Id: 1

CPU time: 3

Period: 20

task 2:

Id: 2

CPU time: 2

Period: 5

task 3:

Id: 3

CPU time: 2

Period: 10

deadline Miss: T_2 at time 5

deadline Miss: T_2 at time 10.

deadline Miss: T_3 at time 10.

deadline Miss: T_2 at time 15.

Gantt Chart:

| T_2 | T_2 | T_3 | T_3 | T_1 | T_2 | T_2 | T_1 | Idle | T_2 | T_3 | T_3 | Idle | T_2 | T_2 | Idle | Idle | Idle |

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20